

# Ajay Venkatesh

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## Summary

Software Engineer with 4+ years of experience building scalable backend systems, serverless platforms, full-stack applications, and cloud-native solutions across fulfillment operations, health analytics, insurance, enterprise consulting, and academic collaboration domains. Proficient in Java, Python, C++, React.js, Node.js, RESTful APIs, AWS, Azure, Docker, and Kubernetes. Experienced in distributed systems, event-driven architectures, CI/CD pipelines, secure APIs, data pipelines, performance optimization, and AI-powered solutions using LLMs, LangChain, RAG, and OpenAI APIs.

## Skills

- Programming Languages:** Java, Python, C++, C#, TypeScript, JavaScript, SQL
- Web Development:** Node.js, Express.js, React.js, HTML, CSS, RESTful APIs, GraphQL APIs, Microservices, Distributed Systems, Event-Driven Architecture, API Development
- Databases and Caching:** PostgreSQL, MySQL, DynamoDB, Redis, Database Design, Query Optimization
- Cloud:** AWS (CDK, Lambda, API Gateway, EC2, S3, DynamoDB, IAM), Azure (OpenAI, Functions, Storage)
- Professional Skills:** Cross-Functional Collaboration, Stakeholder Management, Requirement Analysis, Mentoring, Team Collaboration
- DevOps:** Docker, Kubernetes, CI/CD, GitHub Actions, IaC, Serverless Framework
- AI and Generative AI:** LLMs, RAG, OpenAI APIs, LangChain, NLP, OCR, Prompt Engineering
- Architecture & Integration:** System Design, API Integration, Microservices, Distributed Systems, Event-Driven Architecture, Low-Level Design
- Software Engineering:** Design Patterns, SDLC, Agile Scrum, Code Reviews, Technical Documentation, Mentoring
- Security and Quality:** OAuth, SSO, RBAC, KMS, SSM, Secure Coding, Unit Testing, Integration Testing, Mockito, Debugging

## Experience

### CampusMesh

Software Engineer

Jan 2026 – Present

New York, US

- Built a serverless **Microservices** backend in **Node.js** with **REST APIs**, **AWS Chime SDK**, **DynamoDB**, and **MySQL** to power real-time **chat, video calls, and document sharing** for the study group feature, engaging **10K+** students across **100+** study groups.
- Built a personalized **recommendation engine** using **collaborative filtering** and behavioral analytics to surface relevant study groups and content for **10K+** users, increasing engagement by **28%** and retention by **50%**.
- Standardized serverless deployment pipelines across **4 environments** using **AWS Lambda**, **API Gateway**, and **CDK** with reusable configuration and least-privilege **IAM** to enforce consistent release practices, cutting deployment setup time by **65%**.

### Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Seattle, US

- Designed a **UI console for Amazon FC** using **RBAC** with **OAuth**, **Midway SSO**, and **HELIS** to centralize large-scale **video and image evidence** review with secure role-based access; presented the system design to manager and stakeholders for approval before implementation.
- Provisioned serverless **AWS** infrastructure using **CDK** and **CloudFormation** stacks with **API Gateway**, **Lambda**, **DynamoDB**, and **S3** to enable repeatable multi-region deployments; configured **multi-stage CI/CD pipelines** across dev, staging, and production, reducing provisioning time by **70%**.
- Built a **React 18** frontend with **lazy loading**, **client-side caching**, and **URL-synchronized filter state** via query parameters to enable **deep linking** and shareable filter context across FC teams, reducing page load time by **40%** on media-heavy evidence views.
- Engineered **Java** backend services using **Coral**, **Smithy**, **Dagger**, and **CBOR serialization** with low-level design patterns to optimize request handling and maintain service contracts, reducing execution time by **35%** and ensuring reliability through unit and integration testing with **Mockito**.

### New York University (Novel AI Technologies, Inc.)

Software Developer

Aug 2024 – Present

New York, US

- Architected **Firebeats**, an NSF-funded (\$100K) health app using **React Native**, **Firebase Auth/Firestore**, **AWS Amplify**, and **HealthKit** to unify wearable data (Fitbit, Garmin) and food logging into real-time health dashboards, supporting health monitoring across iOS and Android.
- Engineered health analytics pipelines and **Redux-backed** dashboards using **Node.js**, **PostgreSQL**, and **AWS QuickSight** to process **250K+** health records across activity, sleep, and heart-rate streams, enabling real-time trend tracking and research-ready reporting.
- Developed broker and carrier **insurance portals** using **React**, **Node.js**, **REST APIs**, **AWS Cognito**, and **RBAC** to automate client reporting and row-level access control workflows, generating **\$50K+** in revenue across **6** insurance clients.
- Implemented the **brokerapi** submission lifecycle using **Node.js** and **PostgreSQL** with duplicate guards and policy-year logic to automate underwriting data extraction from form submission to carrier approval, reducing manual processing overhead across 5-year analysis pipelines.

### PricewaterhouseCoopers (PwC)

Senior Software Engineer

Aug 2021 – Aug 2024

Bengaluru, India

- Developed a distributed **Azure** backend integrated with **Microsoft CRM** and **SQL Server** using event-driven processing and fault-tolerant retries to power enterprise healthcare operations, improving workflow reliability and reducing unhandled processing failures.
- Deployed an **NLP-based OCR** model on **Azure** to automate structured document extraction from enterprise health records, cutting manual processing time by **75%** and accelerating downstream reporting cycles.
- Built asynchronous **CRM plugins** and background workflows using **C#**, **Azure Service Bus**, and **SQL Server** to decouple long-running health operations from user-facing flows, improving transaction reliability and reducing processing latency by **40%**.
- Optimized high-volume **data ingestion** pipelines using **Python** parallel processing on **100K-row** Excel feeds to accelerate enterprise analytics delivery, cutting execution time from **5 hours to 1 hour**.
- Led **code reviews** and mentored engineers on software design principles and secure coding practices to elevate delivery quality, reducing production bugs by **25%** across enterprise delivery teams.

## Project

### Advanced Expense Tracker | Java, JavaFX, JDBC, SQLite, JFreeChart, OOP, MVC, DAO

- Architected a modular **expense management** desktop application using **Java**, **JavaFX**, **JDBC**, and **SQLite** with **MVC** and **DAO** patterns to support secure multi-user login, expense tracking, and group cost splitting across shared workflows.
- Implemented **concurrent transaction handling** and real-time cost splitting using **Java multithreading** and synchronized blocks to ensure thread-safe database updates across shared expense workflows, eliminating race conditions under concurrent multi-user load.

## Education

New York University — M.S. in Computer Engineering | GPA: 3.9/4.0

Aug 2024 – May 2026

Relevant Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Visvesvaraya Technological University — B.E. in Electronics and Communication | GPA: 8.9/10

Jun 2017 – Aug 2021